Backyard Hydroponics: A Practical Guide

Hydroponics is a method of growing plants without soil, in which the roots sit directly in a nutrient-rich water solution. Because the plant does not have to search through soil for nutrients, growth rates in a well-tuned hydroponic system are often faster than in traditional garden beds.

The main conclusion of this guide is that a simple deep water culture setup, built from a plastic tote and an aquarium air pump, is the cheapest reliable way for a beginner to start growing leafy greens indoors year round.

Choosing a System

There are several common hydroponic system types: deep water culture, nutrient film technique, ebb and flow, and drip systems. Deep water culture is the easiest to build and maintain for a first-time grower because it has no moving parts other than an air pump.

Nutrient film technique circulates a thin film of nutrient solution over the roots continuously and is popular in commercial lettuce farms, but it requires a reliable pump and is less forgiving of power outages than deep water culture.

Nutrients and pH

Plants grown hydroponically need a balanced nutrient solution containing nitrogen, phosphorus, potassium, and trace minerals such as iron and magnesium. Most hobbyists use a commercial two-part nutrient concentrate mixed into water at the ratio printed on the bottle.

The pH of the nutrient solution should be kept between 5.5 and 6.5 for most leafy greens, since nutrients become harder for roots to absorb outside that range even if the nutrients themselves are present in the water.

Common Problems

Root rot is the most common failure in hydroponic systems and is usually caused by warm, poorly oxygenated water. Keeping the reservoir below 21 degrees Celsius and running the air pump continuously prevents most cases.

Algae growth on sunlit reservoir walls competes with plants for nutrients and can be avoided by using an opaque container or wrapping a clear tote in dark plastic sheeting.